Project Design Phase-I
Solution Architecture

Date	8 Mar 2026
Team ID	PNT2022TMIDxxxxxx
Project Name	Project - Visualization Tool for Electric Vehicle Charge and Range Analysis
Maximum Marks	4 Marks

Solution Architecture

The solution architecture defines how the system collects, processes, analyzes, and visualizes Electric Vehicle (EV) data. The system integrates data storage, data analytics, visualization, and web deployment to provide insights on EV charging patterns, battery performance, and vehicle efficiency.

Architecture Components

1. Data Collection Layer
EV datasets are collected from publicly available sources containing information about EV models, charging stations, battery capacity, price, range, and efficiency.

2. Database Layer
The collected datasets are stored in a SQL database where data can be organized, managed, and queried using SQL operations.

3. Data Preparation Layer
The stored data is cleaned, transformed, and structured to remove inconsistencies, handle missing values, and prepare the dataset for visualization.

4. Data Visualization Layer
The prepared dataset is connected to Tableau to create visualizations such as charts, maps, and graphs. These visualizations help analyze EV charging patterns, brand performance, price comparisons, and vehicle efficiency.

5. Dashboard & Story Layer
Interactive dashboards and data stories are created in Tableau to present insights such as EV charging stations in India, top EV brands, vehicle price comparisons, and efficiency analysis.

6. Web Integration Layer
The Tableau dashboards and stories are embedded into a Flask web application, allowing users to access the visualizations through a web interface.

7. User Interface Layer
Users such as city planners, fleet managers, and consumers interact with the web interface to explore dashboards, apply filters, and analyze EV data.

Example - Solution Architecture Diagram:

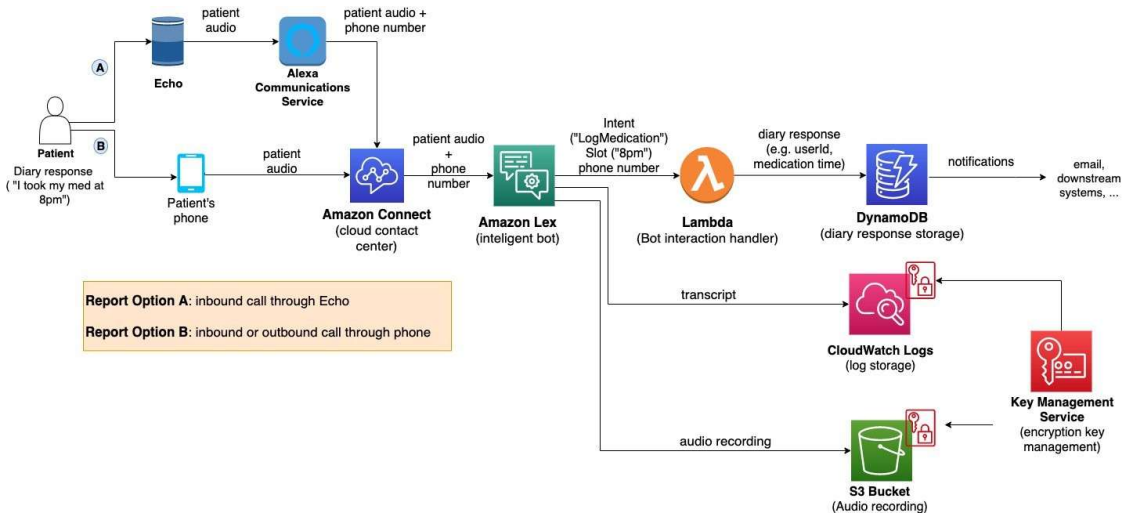

Figure 1: Architecture and data flow of the voice patient diary sample application